

PROBABILITY :-

→ Probability is a measure of likelihood of a particular event happening/occurring.

$\rightarrow$ It Ranges FROM 0 $\longleftrightarrow$ 1
 | └ Certainty
 ^oImpossibility

→ Value in between 0 & 1 like 0.5 and 0.6 Represents varying degree of likelihood.

* SAMPLE SPACE

Set of all possible outcomes of a random event, experiment

Ex:- Tossing a coin, $S = \{T, H\}$

Tossing two coins, $S = \{TT, TH, HT, HH\}$

Rolling / Throwing a Die, $S = \{1, 2, 3, 4, 5, 6\}$

Rolling/Throwing two dies, $S = \{36\}$ elements

★ EVENT

- Event is the subset of the sample space that is denoting specific outcome or combination of outcomes.

- Ex: If we roll a die then: $S = \{1, 2, 3, 4, 5, 6\}$

$A = \text{Getting an even number}$

$$A = \{2, 4, 6\}$$

B = Getting Prime number
 $B = \{2, 3, 5\}$

* PROBABILITY FUNCTION :

- A probability function assign a probability to each event.
- For example :-

$$P(\text{getting a Head}) = \left(\frac{1}{2}\right) = \left(\frac{\text{no. of Favourable outcomes}}{\text{Total no. of outcomes}}\right)$$

$$P(\text{getting even number}) = \left(\frac{3}{6}\right) = \left(\frac{1}{2}\right) = 0.50 = 50\% \text{ chance}$$

* COMPLEMENT OF AN EVENT

- Complement of an event is denoted as A' or A^c , is the set of all outcomes in the sample space that are not in A .
- If A is the event "it rains today", then A' is "it does not rain today".

Mathematically;

$$P(A') = 1 - P(A)$$

Ex :- $S = \{1, 2, 3, 4, 5, 6\}$

$$A = \{2, 4, 6\}$$

$$\therefore A^c = \{1, 3, 5\}$$

$$\therefore \text{If } P(A) = 0.5$$

$$P(A') = 1 - P(A)$$

$$= 0.5$$